This is an excellent collection of package templates that demonstrates deep understanding of both LFS/BLFS and the unique constraints of an offline build system. The templates are generally well-structured, but I found several issues that need attention.

1. Configure Flags

Status: Mostly correct, with some notable omissions and errors

glibc-core — Missing Critical Configure Flags

```
../configure --prefix=/usr               \
   --build=x86_64-igos-linux-gnu         \
   --host=x86_64-igos-linux-gnu          \
   --disable-werror                      \
   --disable-nscd                        \
   libc_cv_slibdir=/usr/lib              \
   --enable-stack-protector=strong       \
   --enable-kernel=5.4
```

Issues:

- Missing --disable-profile (profiling libraries aren't needed)
- Missing --disable-timezone-tools (timezone data is handled separately in post_install — correct)
- --enable-kernel=5.4 is the minimum, but LFS 13.0 recommends --enable-kernel=4.14 or higher. 5.4 is fine.
- No --enable-bind-now (security hardening). Consider adding.

nss — Missing Critical Build Flags

```
make BUILD_OPT=1                    \
    NSPR_INCLUDE_DIR=/usr/include/nspr  \
    USE_SYSTEM_ZLIB=1               \
    ZLIB_LIBS=-lz                   \
    NSS_ENABLE_WERROR=0             \
    NSS_USE_SYSTEM_SQLITE=1         \
    $([ $(uname -m) = x86_64 ] && echo USE_64=1) \
```

```
    -j${IGOS_JOBS}
```

Issues:

- Missing NSDISTMODE=copy — Without this, nss may create symlinks instead of copying files, causing issues with DESTDIR staging.
- Missing NSS_DISABLE_GTESTS=1 — The test suite won't run without additional dependencies (not critical).
- The USE_64=1 flag is needed for x86_64, but you're not setting USE_32=1 for 32-bit. That's fine.

openssh — Missing Critical Options

```
./configure --prefix=/usr                    \
        --sysconfdir=/etc/ssh                \
        --with-privsep-path=/var/lib/sshd    \
        --with-default-path=/usr/bin         \
        --with-superuser-path=/usr/sbin:/usr/bin \
        --with-pid-dir=/run                  \
        --with-pam
```

Issues:

- Missing --with-md5-passwords — Needed for compatibility with some PAM configurations.
- Missing --without-openssl-header-check — Some OpenSSL versions trigger false positives.
- Consider --with-kerberos5 if you plan to support Kerberos authentication.

mesa — Incorrect Vulkan Driver Selection

```
-D vulkan-drivers=auto
```

auto includes all Vulkan drivers, which adds significant compile time. For a desktop system, you likely only need intel,amd (or intel,amd,swrast). Specify explicitly.

rust — Bootstrap Tarball Path Issue

```
cp -v "${IGOS_SOURCES}/rustc-1.92.0-x86_64-unknown-linux-gnu.tar.xz" \
    build/cache/2025-12-11/
```

The date directory 2025-12-11 is hardcoded from the bootstrap compiler's release date. When the bootstrap compiler updates, this will break. Use a variable or detect from the tarball:

```
bootstrap_date=$(tar tf "${IGOS_SOURCES}/rustc-1.92.0-x86_64-unknown-linux-gnu.tar.xz" | head -1 | cut -d/ -f1)
mkdir -p "build/cache/${bootstrap_date}"
```

vala — Incorrect Make Target

```
make bootstrap -j${IGOS_JOBS}
```

BLFS 13.0 uses make (not make bootstrap) for vala. The bootstrap target may not exist in all versions.

webkitgtk — Missing Critical CMake Flags

```
-D USE_WOFF2=OFF
```

WOFF2 is required for web fonts. Setting this to OFF will break web font rendering in many websites. BLFS 13.0 enables it. Either add WOFF2 to dependencies or remove this flag.

gvfs — Missing Backend Dependencies

The pass1 template includes all backends except google and the pass2-specific ones. However, it doesn't include libimobiledevice in dependencies despite having it in the YAML. The YAML lists it, so that's correct.

2. Dependency Declarations

Status: Generally good, but missing critical dependencies

glibc-core — Missing Build Dependencies

The YAML shows empty dependencies, which is correct for a bootstrap package. However, glibc's post_install uses tar and zic (from tzdata) — these are runtime dependencies, not build deps.

nss — Missing Runtime Dependency

```
runtime:
  - p11-kit
```

Correct — the p11-kit trust module is linked. But missing sqlite (runtime dep for NSS database).

mesa — Missing Runtime Dependencies

```
runtime: []
```

Mesa at runtime requires:

- libdrm (already in build deps, but should be runtime too)
- elfutils (for debug symbols)
- libxshmfence (for X11 synchronization)

rust — Missing Build Dependencies

The YAML lists llvm, cmake, curl as build deps. Missing:

- python3 (required for build scripts)
- git (some build scripts use it for version detection)
- perl (for some test scripts)

webkitgtk — Missing Build Dependencies

```
build:
  - webkitgtk-gtk3  # This creates a circular dependency!
```

webkitgtk (GTK4 version) depends on webkitgtk-gtk3? That's incorrect. They are parallel-installable versions of the same library, not dependencies of each other. The GTK4 version should NOT depend on the GTK3 version.

gvfs-pass2 — Incorrect Dependency Chain

```
dependencies:
  build:
  - gvfs  # This creates a circular dependency
```

gvfs-pass2 should not depend on gvfs. The two builds are separate; pass2 rebuilds the package, it doesn't depend on pass1. The correct approach is to have pass2 installed **after** pass1, not depend on it. Remove gvfs from build deps.

argcomplete — Missing Build Dependency

```
build:
  - hatchling
```

Correct, but missing python3 in build deps (assumed to be available).

3. DESTDIR Handling

Status: Excellent — all packages correctly use DESTDIR

All templates properly use DESTDIR="$DESTDIR" in their install commands. Special mentions:

- **glibc-core**: Uses install_root="$DESTDIR" instead of DESTDIR — correct for glibc.
- **nss**: Manual installation to $DESTDIR — correct.
- **rust**: Uses DESTDIR="$DESTDIR" ./x.py install — correct.

- **vala**: Uses make DESTDIR="$DESTDIR" install — correct.

One issue: In openssh's do_install, you install a systemd unit from /mnt/intergenos/config/. This path assumes the build system's source tree is mounted at /mnt/intergenos inside the chroot. In your setup, that's correct (virtiofs mount). But document this assumption.

4. Missing post_install() Hooks

Status: Several packages missing critical post_install steps

mesa — Missing post_install

Mesa requires:

- ldconfig (to update library cache)
- update-alternatives for OpenGL if multiple implementations exist
- glib-compile-schemas for GSettings (if any schemas installed)

gtk4 — Missing post_install

GTK4 installs GSettings schemas, GDBus interfaces, and icon caches:

```
post_install() {
  glib-compile-schemas /usr/share/glib-2.0/schemas
  gtk-query-immodules-4.0 --update-cache
}
```

gst-plugins-base and gst-plugins-base-pass2 — Missing post_install

GStreamer plugins need gst-update-registry:

```
post_install() {
  gst-update-registry
}
```

libmtp — Missing udev rules

libmtp installs udev rules that need to be triggered:

```
post_install() {
  udevadm control --reload-rules
  udevadm trigger
}
```

libgphoto2 — Missing udev rules and hotplug

```
post_install() {
  udevadm control --reload-rules
  udevadm trigger
  # Optional: restart gvfs-gphoto2-volume-monitor
}
```

libimobiledevice — Missing udev rules and usbmuxd activation

```
post_install() {
  udevadm control --reload-rules
  udevadm trigger
  systemctl enable usbmuxd.service 2>/dev/null || true
}
```

libbluray — Missing font cache update

```
post_install() {
  fc-cache -v
}
```

gnome-keyring — Missing PAM configuration and daemon activation

The BLFS book specifies installing PAM config files and enabling the keyring daemon.

ibus — Missing input method registration

```
post_install() {
   # Already has glib-compile-schemas, but missing:
   ibus-daemon --xim &
   # Or at least document that user must run ibus-setup
}
```

mutter — Missing session configuration

Mutter as a Wayland compositor needs to be registered as a session type.

5. Offline Build Compatibility

Status: Good, with a few violations

rust — Attempts Network Access Despite Pre-downloaded Bootstraps

The configure() function copies bootstrap tarballs to build/cache/2025-12-11/, which prevents x.py from downloading them. However, the bootstrap.toml has locked-deps = true but missing download-rustc = false. Add:

```
[rust]
download-rustc = false
```

shaderc — No Offline Mode

shaderc's build system (CMake) may attempt to download googletest and other dependencies. The template doesn't disable this. Add:

```
configure() {
   # Disable downloading during build
   sed -i '/add_subdirectory(third_party)/d' CMakeLists.txt
   # ... rest of configure
}
```

argcomplete — Uses pip Without Offline Mode

pip3 will attempt to connect to PyPI if dependencies are missing. The template uses --no-index --no-deps, which is correct. However, pip itself may still try to fetch package metadata. Add:

```
build() {
   pip3 wheel -w dist --no-build-isolation --no-deps --no-cache-dir --no-index $PWD
}
```

glycin — Offline Rust Crates Work

The template correctly extracts glycin-2.0.8-vendor.tar.gz and configures cargo to use vendored sources. However, the patch application after vendor extraction may invalidate cargo checksums. The template clears them with:

```
for cs in vendor/*/.cargo-checksum.json; do
   sed -i 's/"files":{[^}]*}/"files":{}/' "$cs"
done
```

This is correct but aggressive. Consider only clearing checksums for patched crates.

ibus — Downloads UCD.zip During Configure

The template checks for /usr/share/unicode/ucd/NamesList.txt and extracts from IGOS_SOURCES if present. This is correct for offline builds, but it's extracting to /usr/share/unicode/ucd **during configure**, which modifies the live system (even with DESTDIR). This should be moved to post_install.

6. Two-Pass Builds

Status: Correct wiring, but missing rebuild triggers

freetype2 (not in this set, but pattern established)

The two-pass pattern is documented but not shown. The critical point is that pass2 must rebuild **after** its additional dependencies. In the build order, this is handled by the dependency graph.

gst-plugins-base-pass2

The template correctly adds mesa as a build dependency. The build order will be:

1. gst-plugins-base (pass1, no mesa)
2. mesa (provides GL)
3. gst-plugins-base-pass2 (rebuild with GL)

Missing: The pass2 template doesn't remove the pass1 installation. This means files from pass1 will be overwritten. That's acceptable if pass2 is installed **after** pass1 (which it will be), but there's a risk of leftover files. Consider having pass2 explicitly remove the pass1 package first, or ensure the build system handles replacement.

gvfs-pass2

Similar pattern, but the dependency should be on gnome-online-accounts and libmsgraph, not on gvfs itself. The YAML currently has:

```
dependencies:
  build:
  - gvfs
  - gnome-online-accounts
  - libmsgraph
```

Remove gvfs from build deps. The correct dependency is that pass2 must be built **after** goa and libmsgraph, but it doesn't need pass1 installed to build.

Critical Issues Summary

Package	Issue	Severity

glibc-core	Missing hardening flags	Medium
nss	Missing NSDISTMODE=copy	High — breaks DESTDIR
mesa	Vulkan drivers set to auto (too many)	Low
rust	Hardcoded bootstrap date	Medium
webkitgtk	USE_WOFF2=OFF breaks web fonts	High
webkitgtk	Circular dependency on webkitgtk-gtk3	High
gvfs-pass2	Circular dependency on gvfs	High
mesa	Missing post_install (ldconfig, schemas)	Medium
gtk4	Missing post_install (schemas, immodules)	Medium
libmtp/libgphoto2	Missing udev reload	Medium
ibus	Extracts UCD to live system during configure	High
rust	Missing download-rustc = false	High
shaderc	May attempt network downloads	Medium

Security Issues

openssh — Hardcoded Root Login

sed -i 's/^#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config

Enabling root login with password is a security risk. The installer sets a root password, but this should be PermitRootLogin prohibit-password (key-only) or without-password. Consider making this configurable.

glibc — Hardcoded Timezone

```
zic -d $ZONEINFO -p America/New_York
```

Hardcoding America/New_York is not appropriate for a distribution. The installer should set the timezone, and the post_install should use that value.

rust — Unrestricted PATH in profile.d

```
cat > /etc/profile.d/rustc.sh << "PROFILE"
pathprepend /opt/rustc/bin        PATH
PROFILE
```

The pathprepend function may not be defined. This will cause errors in non-bash shells. Use standard PATH manipulation:

```
cat > /etc/profile.d/rustc.sh << "PROFILE"
export PATH="/opt/rustc/bin:${PATH}"
PROFILE
```

Recommendations

1. **Add a validation script** that checks each package.yml for:
    a. Required fields (name, version, source, etc.)
    b. Valid build_style
    c. No circular dependencies
    d. All dependencies exist
2. **Standardize post_install hooks** for common operations:
    a. ldconfig for libraries
    b. glib-compile-schemas for GSettings
    c. gtk-query-immodules for GTK input methods
    d. fc-cache for fonts
    e. systemctl enable for services
    f. udevadm for device rules
3. **Add a pre_configure phase** for operations that need to run before configure (like extracting vendored crates). The glycin template uses this pattern — standardize it.

4. **Document the offline build assumptions**:
    a. Which packages need pre-vendored crates (glycin, rust, mesa)
    b. Which packages need pre-downloaded bootstrap binaries (rust)
    c. Where to place these additional tarballs
5. **Create a package template validator** that checks for common mistakes:
    a. Missing DESTDIR in install commands
    b. Hardcoded paths outside $DESTDIR
    c. Network downloads in configure/build

These templates represent a significant amount of careful work. The issues I've found are fixable, and the overall quality is high. With the corrections above, especially the circular dependencies and missing DESTDIR flags, this will be a solid foundation for the distribution.